

PRE-CLASS READING 4 OF 5

Using AI to Accelerate Scientific Discovery — and AlphaFold 3

Demis Hassabis (2023), EPFL campus lecture. Abramson, J. et al. (2024), *Nature* 630, 493–500.

ASSIGNED READING

Campus lecture — EPFL, 26 May 2023 — skip Q&A if short on time

~40 min

Paper — abstract + skim main figures

~15 min

≈ 55 minutes

Note on citation. Your syllabus lists this as a “TED talk.” It isn't one — it's Demis Hassabis's Dr. Honoris Causa campus lecture at EPFL, 26 May 2023. A differently-titled talk exists on TED (“How AI is unlocking the secrets of nature and the universe,” TED2024), but it's a different, shorter, more forward-looking piece. This reading uses the EPFL lecture because its title and date match what the course intended.

KEY TAKEAWAY

AlphaFold didn't just improve on existing protein-structure tools — it closed a fifty-year-old open problem in biology outright, in a fraction of the time and cost of experimental methods. Hassabis frames this as a template: pair a hard, well-defined scientific problem with the right AI architecture, and you can compress decades of progress into years.

WHY IT MATTERS FOR THIS COURSE

This is your case study for “AI + domain expertise is multiplicative, not additive” — a biologist alone couldn't predict structures at this scale; AlphaFold alone doesn't know which predictions matter biologically. Together, they solved a Nobel-worthy problem (Hassabis and John Jumper won the 2024 Nobel Prize in Chemistry for this work). The lecture also gives you Hassabis's own founding story for Isomorphic Labs — a direct example of a research breakthrough becoming a company, relevant when you scope your own end-of-course project.

MAIN POINTS

The lecture

- **Games as a training ground, not the destination.** Hassabis traces DeepMind's arc from AlphaGo (2016) — the first program to beat a world champion at Go — to argue that games were always a proving ground for general learning methods, not the end goal. The real target was always applying the same underlying techniques to real-world scientific problems.
- **Protein folding: a fifty-year grand challenge.** Proteins are defined by their amino-acid sequence, but what they *do* depends on the 3D shape they fold into — and predicting that shape from sequence alone had resisted biology since the 1970s. AlphaFold (2020) solved this at CASP14, the field's blind-test competition, at a level organizers described as solving the problem.
- **Speed and scale as the real story.** Once solved, the same model could predict a structure in seconds rather than the months or years experimental methods (X-ray crystallography, cryo-EM) require. This is why DeepMind could release predicted structures for essentially all ~200 million proteins known to science — a task that would have taken the experimental pipeline centuries.
- **From breakthrough to company.** The AlphaFold result led Hassabis to found Isomorphic Labs (2021) specifically to apply the same AI methods to drug discovery — predicting not just protein structures but how drug molecules bind to them. This is the lecture's throughline: a scientific result becomes a venture once you see the downstream application.
- **General lesson for this course:** pick a problem where (a) the underlying pattern is learnable from data, and (b) getting the answer faster or cheaper unlocks a chain of further work. That's the same test you should apply when scoping your own end-of-course project.

The paper — what AlphaFold 3 actually adds

- **Beyond single proteins.** AlphaFold 2 predicted protein structure. AlphaFold 3 predicts the *joint* structure of complexes — proteins together with nucleic acids (DNA/RNA), small molecules, ions, and chemically modified residues — within one unified deep-learning architecture, rather than needing a different specialized tool per interaction type.

“The new AlphaFold model demonstrates substantially improved accuracy over many previous specialized tools: far greater accuracy for protein–ligand interactions compared with state-of-the-art docking tools, much higher accuracy for protein–nucleic acid interactions compared with nucleic-acid-specific predictors and substantially higher antibody–antigen prediction accuracy compared with AlphaFold-Multimer v.2.3.”

- **Beating dedicated tools at their own job.** On the PoseBusters benchmark (428 recent protein–ligand structures), AlphaFold 3 outperforms classical docking software like Vina — even though it uses no structural inputs the docking tools rely on (statistically significant at $P = 2.27 \times 10^{-13}$). It also outperforms other blind-docking AI models such as RoseTTAFold All-Atom.
- **Not perfect — has known failure modes.** The paper itself reports a 4.4% chirality-violation rate on PoseBusters (occasionally predicting a molecule's mirror image) and weaker performance on complex glycan structures (~42–46% success). On CASP15 RNA targets, AlphaFold 3 beat other AI-based methods but still fell short of the best human-expert-assisted submission. Worth holding onto: even a landmark result has an honestly reported jagged frontier of its own.
- **Open access, now with open code.** DeepMind provided free non-commercial access via the AlphaFold Server from launch. A November 2024 addendum (not a correction to the results) announced release of the underlying inference code on GitHub — meaning by 2026 the model is both usable and inspectable, not just a black-box API.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here — these are meant to surface how you think, not to test recall.

1. Hassabis describes games like Go as a “training ground” — not the real goal, but a safe, controlled place to build and test a general skill before pointing it at something messier and real. Think of a time in your own studies or life when a skill built in one narrow setting (a class exercise, a game, a training simulation) turned out to transfer somewhere you didn't expect. What made it transferable — and what wouldn't have transferred?
2. AlphaFold 3 replaced a collection of specialized tools with one general-purpose model. You've likely lived through a version of this yourself — a phone that absorbed what used to be a separate camera, music player and map, for instance. What was gained by combining them into one general tool? Was anything lost?
3. The AlphaFold 3 paper reports its own weaknesses — errors, cases where it underperforms — in the very same document that reports its breakthroughs. You're about to spend two years reading confident claims about AI, from papers, vendors, and eventually employers. What in this paper made its claims feel more trustworthy to you? What's one thing you'll now look for when a claim *doesn't* show its own weak spots?

SOURCE LINKS

Lecture · [EPFL Memento — Campus Lecture with Demis Hassabis](#) · video via [Breakthrough Prize](#)

Paper · [Nature — Accurate structure prediction of biomolecular interactions with AlphaFold 3](#) · DOI: 10.1038/s41586-024-07487-w

Addendum (code release) · [Nature — Addendum](#)

AlphaFold Server (free, non-commercial) · alphafoldserver.com

Inference code · github.com/google-deepmind/alphafold3

CAVEATS

1. **This summary is not a substitute for the reading.** It is a map, not the territory. Watch the lecture and read the paper's abstract and figures directly — the paper in particular is written for a specialist audience, and this summary simplifies its statistics.
2. **This summary was generated and fact-checked using Anthropic models,** including direct retrieval of the paper's abstract, DOI, and volume/page citation from Nature.com, and verification of the lecture's actual title, date, and format against EPFL's own event records. Errors are still possible — the original sources are the authority. Flag anything that looks wrong.